

# MRUNALINI RAMNATH

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## EDUCATION

### Carnegie Mellon University

Master of Science, Artificial Intelligence & Innovation

Pittsburgh, PA

Expected May 2027

**Selected Coursework:** Machine Learning (PhD), Deep Learning, AI Engineering, Adv. Natural Language Processing, Multimodal Machine Learning, Computational Data Science, Data Structures and Algorithms

### Birla Institute of Technology & Science, Pilani

Bachelor of Engineering, Electronics & Communication Engineering; Master of Science, Physics

Hyderabad, India

Jun 2022

**Selected Coursework:** Artificial Intelligence, Object-Oriented Programming, Operating Systems, Digital Image Processing

## WORK EXPERIENCE

### American Express

Bengaluru, India

Senior Analyst

Feb 2024– Jun 2025

- **ML Pipeline Migration:** Led migration of large-scale production ML feature pipelines from IBM mainframe to a proprietary cloud platform for online credit-risk decisioning; executed canary deployments and sustained 99.9% uptime.
- **Data Reliability:** Built lineage-aware ingestion and validation workflows to enforce feature freshness, access governance, and dependency integrity in production ML pipelines.
- **Feature Compression:** Used feature selection and clustering to compress feature space by 43%, boosting throughput.

Analyst

Jul 2022– Jan 2024

- **Feature Optimization:** Cut feature usage 32% via tree-based selection, correlation filtering, and mutual information.
- **QA Automation:** Built PySpark/HiveQL validation framework for production ML datasets, saving 2500+ QA hours/quarter.
- **Business Impact:** Scaled feature generation pipelines driving \$15M/year impact.

Analyst Intern

Jan 2022– Jun 2022

- **Alert Reporting:** Automated triggered-alert retrieval and reporting, saving 1000+ QA hours/quarter.
- **False Alerts:** Reduced false-alert rate 13% by improving feature logic and alert filtering.

### Adobe

Bengaluru, India | Remote

Software Development Intern

May 2021– Jul 2021

- **Vulnerability Review:** Analyzed and enhanced 10+ Kubernetes clusters; documented 15+ critical security gaps.
- **Benchmarking:** Evaluated kube-bench/kube-hunter/AquaSecurity/kube-score/kubesec, improving coverage 26%.
- **Remediation Plan:** Proposed remediations that cut monitoring overhead 15% with seamless integration.

## ACADEMIC PROJECTS

### Fine-Tuning a Step-Efficient Multimodal Web Agent

Pittsburgh, PA

- **Behavior Fine-Tuning:** Fine-tuning a candidate-ranking multimodal agent on Mind2Web for grounded action selection.
- **Step Optimization:** Preference-training in VisualWebArena to minimize steps and decision errors.

### Privacy-Compliant LLM Feature Engineering for Zulip

Pittsburgh, PA

- **LLM Integration:** Engineered chat and inbox summarization, with topic-drift detection in the Zulip open-source codebase.
- **Privacy-Safe Inference:** Implemented identity-masking with deterministic re-mapping to prevent sensitive data exposure.
- **Inference Efficiency:** Optimized token usage and latency via trigger-based invocation, rolling context windows, and message batching. Achieved sub-1-minute inference.
- **Clean Outputs:** Citation-backed summaries with structured prompts; constraints to improve reliability, reduce injection risk.

### Efficient Multimodal Fusion with Bottleneck Transformers

Pittsburgh, PA

- **Fusion Architectures:** Implemented early, mid, late, and bottleneck fusion variants on CMU MOSI/MOSEI.
- **Accuracy Gains:** Designed hierarchical bottleneck fusion: +4.3% Acc2 and +2.8% F1 vs. strongest baseline.
- **Compute Tradeoffs:** Analyzed accuracy-compute behavior through FLOPs-aware ablations.

## RESEARCH PROJECTS

### Computer Vision Lab, University of Ljubljana

Ljubljana, Slovenia | Remote

Machine Learning Research Intern

Jun 2021– Dec 2021

- **Segmentation Module:** Integrated CNN segmentation into a biometric eye-ID multi-step model pipeline.
- **Model Benchmarking:** Benchmarked U-Net, MobileNet, and DenseNet variants for vascular-feature identification.
- **Latency Reduction:** Cut inference latency to 27% via a single compact model with no accuracy regression.

## SKILLS

**Languages:** Python, C, C++, SQL, HiveQL, MATLAB, TypeScript, HTML/CSS, Git

**ML Libraries:** PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, PySpark, Apache Spark

**Machine Learning:** Multimodal Models, LLMs, LMMs, VLMs, Computer Vision, Training/Optimization, Fine-Tuning, LoRA

**Frameworks & Libraries:** HuggingFace Transformers, OpenCV, NLTK, Django, Flask

**Cloud & Platforms:** AWS, Azure, Kubernetes, Docker